Alkanes, Alkenes, and Alkynes

Ingyu Bahng

May 2026

Contents

1 Alkenes

Core Concept 1.1: Alkene Orbital Structure

Core Concept 1.2: Alkene Bond Strength

Core Concept 1.3: Relative Stability of Alkenes

1.1 Alkene Synthesis

Core Concept 1.4: Elimination

Core Concept 1.5: Alkenes from ROH By Dehydration

1.2 Alkene Addition Reactions

1.2.1 Catalytic Hydrogenation

Core Concept 1.6: Catalytic Hydrogenation

1.2.2 Electrophilic Additions

Core Concept 1.7: Hydrohalogenation

Core Concept 1.8: Markovnikov Hydration

Core Concept 1.9: Halogenation

Core Concept 1.10: Oxymercuration–Demercuration

1.2.3 Hydroboration-Oxidations

Core Concept 1.11: Hydroboration-Oxidations

1.2.4 Electrophilic Carbene Additions

Core Concept 1.12: Electrophilic Carbene Additions

1.2.5 Electrophile Oxidation

Core Concept 1.13: Peroxycarboxylic Acids

Core Concept 1.14: OsO_4 Syn-Dihydroxylation

Core Concept 1.15: Ozonolysis

1.2.6 Radical Hydrobromination

Core Concept 1.16: Radical Hydrobromination

2 Alkynes

Core Concept 2.1: Alkyne Orbital Structure

Core Concept 2.2: Triple Bond Strength

Core Concept 2.3: Relative Acidity of Alkynes

2.1 Alkyne Synthesis

Core Concept 2.4: Elimination E2 From Dihaloalkanes

Core Concept 2.5: Alkylation of Alkynyl Metals

2.2 Alkyne Reactions

2.2.1 Reductions

Core Concept 2.6: Complete Hydrogenation

2.2.2 Electrophilic Additions (Markovnikov)

Core Concept 2.7: HX Anti Addition

Core Concept 2.8: X_2 Anti Addition

Core Concept 2.9: Mercury Catalyzed Markovnikov Hydration

Core Concept 2.10: Radical HBr Anti-Markovnikov Addition

Core Concept 2.11: Hydroboration-Oxidation